



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO V

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

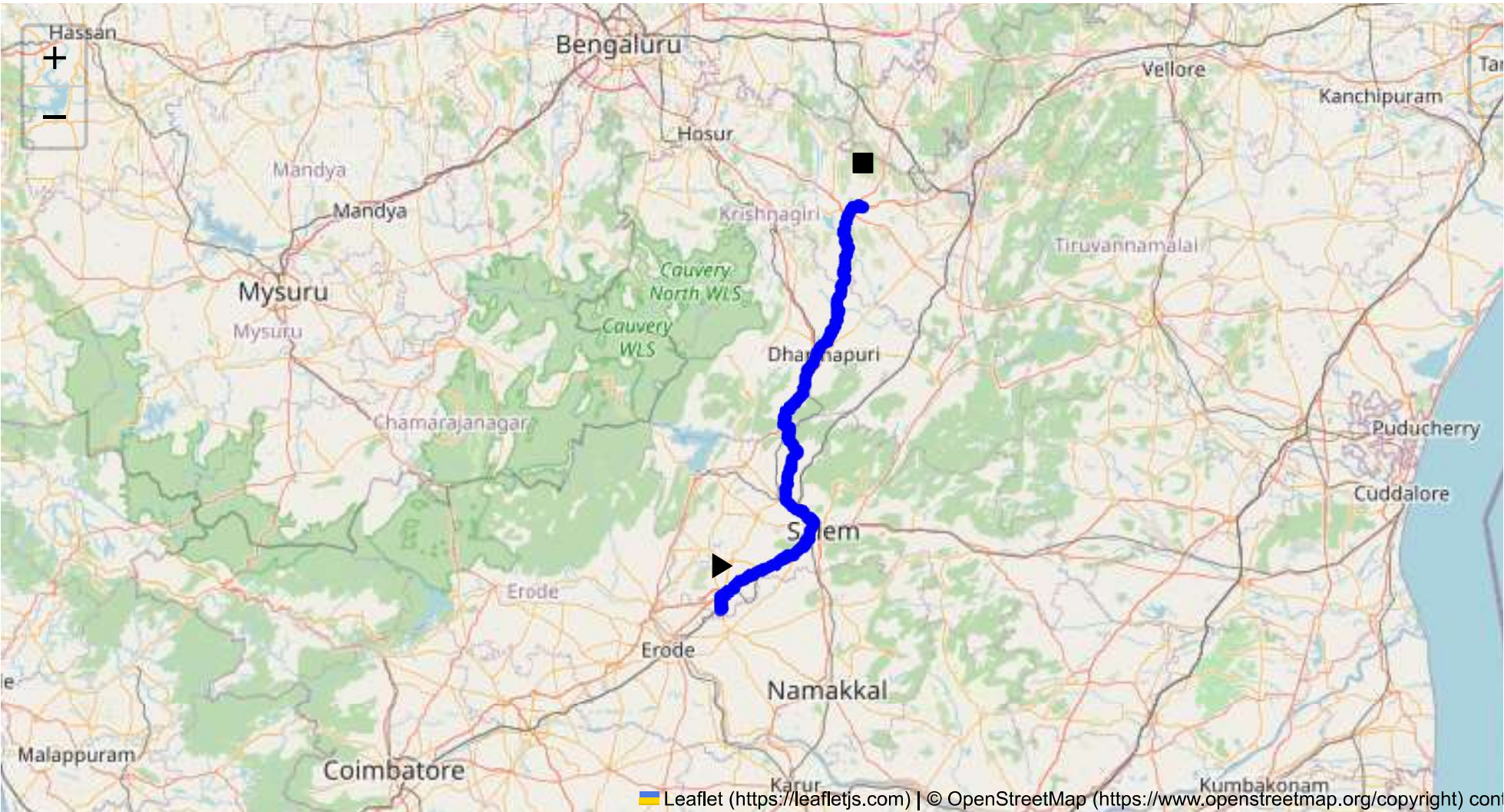
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 157.45 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours
Start: (11.4381, 77.8734)
End: (12.534787, 78.260195)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to G7M6+W33, Krishnagiri, spans approximately 157.45 kilometers, primarily along National Highway 544 (NH544) and NH44. The journey typically takes around 2.81 hours for heavy vehicles carrying hazardous materials, passing through key towns including Dharmapuri and significant stretches of open highway.

2. Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate. The route is susceptible to heavy rainfall from the Southwest and Northeast monsoons (June-November), potentially causing temporary flooding in low-lying areas. High temperatures during summer (March-May), touching up to 40°C, can affect vehicle performance. Wind gusts during monsoon storms may also impact high-sided vehicles.

3. Traffic Patterns

Traffic congestion is commonly observed during morning (7-9 AM) and evening (5-8 PM) peak hours around urban centers like Salem and Dharmapuri. NH544 and NH44, being major highways, may experience delays due to sporadic congestion, especially around toll plazas and major intersections.

4. Road Quality and Infrastructure

Overall, the road quality along NH544 and NH44 is generally good, with well-maintained surfaces. However, watch for occasional potholes and uneven surfaces, particularly after heavy rains. Infrastructure is robust with adequate signage, but construction zones can cause abrupt detours.

5. Alternative Routes for Emergencies

In case of severe congestion or roadblocks, consider taking State Highway routes intersecting NH544 and NH44. SH17 via Thoppur is one alternative that reconnects with NH44 towards Krishnagiri, though it might increase travel time.

6. Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials in Tamil Nadu requires adherence to the Petroleum Rules, 2002, and the Motor Vehicles Act, 1988. Checkpoint inspections are routine; documents related to material nature, safety protocols, and emergency procedures should be up-to-date.

7. Historical Incidents

There have been relatively few high-profile incidents involving heavy vehicles transporting hazardous materials on this route. However, vigilance is advised around heavy vehicle accident-prone zones, especially near major urban turnpikes.

8. Environmental Considerations and Sensitive Areas

The route passes near environmentally sensitive zones like reserve forests around Dharmapuri. Any hazardous spill could severely impact local flora and fauna. Cross-border wildlife activity can occasionally invade roadways.

9. Communication Coverage

Communication infrastructure is well-developed with consistent cellular network coverage along NH544 and NH44. Some isolated stretches between urban centers may experience brief dead zones.

10. Emergency Response Times

Emergency response in urban centers like Salem and Krishnagiri is relatively swift, estimated at 15-20 minutes. For rural stretches, response times can range from 30-45 minutes due to distance from services and challenging terrain.

11. Overall Summary of Risk Assessment

The route is generally safe and efficient for transporting hazardous materials, subject to observance of proper timings to avoid peak hour congestion and attention to severe weather alerts. Infrastructure supports heavy vehicles, with robust emergency services in urban areas. Nonetheless, drivers should remain vigilant for environmental risks and ensure regulatory compliance to mitigate actual and potential hazards. Regular coordination with transportation management authorities is advised for real-time updates and assistance.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	12.5355669, 78.26737299999999	10 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	High	11.49149, 77.89601	15 KM/Hr
8	Turn	High	11.49224, 77.89601	15 KM/Hr
9	Turn	Medium	11.53512, 77.95717	30 KM/Hr
10	Turn	Medium	11.53606, 77.95869	30 KM/Hr
11	Turn	Medium	11.62403, 78.10893	30 KM/Hr
12	Turn	High	11.64183, 78.11810	15 KM/Hr
13	Turn	Medium	11.64338, 78.11998	30 KM/Hr
14	Turn	Medium	11.67903, 78.12796	30 KM/Hr
15	Turn	Medium	11.73968, 78.05098	30 KM/Hr
16	Turn	Medium	11.86790, 78.08347	30 KM/Hr
17	Turn	Medium	11.91357, 78.05868	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
18	Turn	Medium	11.92556, 78.06499	30 KM/Hr
19	Turn	Medium	11.93694, 78.05425	30 KM/Hr
20	Turn	Medium	11.95531, 78.04698	30 KM/Hr
21	Turn	High	11.95803, 78.04964	15 KM/Hr
22	Turn	Medium	11.96068, 78.04960	30 KM/Hr
23	Turn	High	11.96370, 78.05142	15 KM/Hr
24	Turn	High	11.96542, 78.04938	15 KM/Hr
25	Turn	High	11.96650, 78.04927	15 KM/Hr
26	Turn	Medium	12.22422, 78.19127	30 KM/Hr
27	Turn	Medium	12.24685, 78.19561	30 KM/Hr
28	Turn	Medium	12.25012, 78.19946	30 KM/Hr
29	Turn	Medium	12.25280, 78.20041	30 KM/Hr
30	Turn	Medium	12.29065, 78.19892	30 KM/Hr
31	Turn	Medium	12.32012, 78.20734	30 KM/Hr
32	Turn	Medium	12.40099, 78.21805	30 KM/Hr
33	Turn	High	12.50736, 78.22244	15 KM/Hr
34	Blind Spot	Blind Spot	12.53557, 78.26737	10 KM/Hr
35	Turn	High	12.53525, 78.26050	15 KM/Hr
36	Turn	High	12.53499, 78.26032	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
21	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
22	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
24	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
25	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
26	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
27	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
28	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
29	hospital	Global Medical Centre Jagirammapalayam	11.6894409, 78.1108847	30 km/h	Medium
30	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
34	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
35	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
38	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
39	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
40	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
42	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
47	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
31	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
32	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
33	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
36	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium